

Position, Not Capability: How Stock Markets Price Large Language Model Releases

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Abstract

Do stock markets price the capability of large language models, or the economic position of firms around a release? Using 124 day-verified model releases (2022–2026), benchmark capability scores, and double-blind-coded ecosystem positions for Nasdaq-100 firms, we find capability is not positively priced: the slope is zero pooled, significantly negative among closed-source flagships, and zero among unrelated firms. Position is priced: hardware suppliers and rival developers earn 1.6–1.7 percentage points over twenty days, with immediate abnormal volume, while firms deploying AI in conventional businesses show precisely zero price and volume response. Capability matters only as an amplifier of the hardware premium.

Keywords: Large language models, Event study, Capability pricing, AI supply chain, Abnormal volume

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1. Introduction

Large language model (LLM) releases have become routine, salient information events—124 benchmark-qualified releases in our 2022–2026 sample, roughly one per week by the sample’s end. Each arrives with a scorecard: third-party evaluators publish standardized capability indices within days, and financial commentary routinely reads those scores as value-relevant news. A natural hypothesis, implicit in exposure-based studies of AI and asset prices (Eisfeldt et al., 2023; Babina et al., 2024), is that markets reward measured capability: the more capable the model, the larger the revaluation of exposed firms. The hypothesis has intuitive appeal—capability is the technological object a release delivers—and it underlies both scorecard journalism and the growing practice of ranking firms by AI exposure.

This letter tests the capability hypothesis directly and rejects it. Across 87 language-model releases with capability scores, the cross-sectional slope of announcement returns on capability is zero. Where theory says capability rents should be steepest—closed-source releases, whose developers can appropriate capability through API pricing and ecosystem lock-in (Teece, 1986)—the slope is significantly *negative*. What markets price instead, consistently and increasingly, is *where a firm sits* in the AI production chain: suppliers of compute inputs and rival model developers revalue on release, firms that merely use AI do not, and capability matters only by scaling the compute suppliers’ premium.

Our design combines three measurement inputs that are, to our knowledge, new in combination. First, release dates verified to the day against first-party sources—vendor engineering blogs, release notes, model reposi-

ries, and archived first captures—for every event; secondary aggregators, we find, misdate a substantial share of releases, sometimes by months, and misdated events mechanically decouple returns from announcements. Second, a direct measure of model capability: the Artificial Analysis intelligence index, a composite of standardized reasoning and knowledge benchmarks published by a third-party evaluator, rather than a text- or investment-based proxy. Third, ecosystem positions assigned to each (firm, developer) pair by two independent coders under a fixed eight-dimension codebook—hardware supplier, cloud platform, integrator, deployer, enabler, competitor, investor, owner—with 99.0% cell-level agreement and dimension-level Cohen’s κ between 0.84 and 1.00. The firm universe is the Nasdaq-100, so selection into the sample is exogenous to any AI hypothesis, and the thirty-one constituents with no classifiable AI relationship (consumer staples, utilities, health care) supply a within-event control group.

Three results follow. First, capability is not positively priced. The pooled slope of twenty-day cumulative abnormal returns (CARs) on capability is -0.01 pp per index point ($p = 0.51$). Among the fifty closed-source releases the slope is -0.06 pp per point—about -0.8 pp per standard deviation of capability ($p = 0.024$), and -1.3 pp per standard deviation under a three-factor benchmark ($p < 0.001$). Among firms with no ecosystem relationship the slope is an exact zero ($p = 0.80$), as a placebo requires. The negative flagship slope is an anticipation gradient rather than a penalty on capability: the most capable releases are the most rumored and scheduled, so their information reaches prices before the announcement window, and the highest-scoring closed releases in the sample all carry *positive* average CARs. Second, posi-

tion is priced. Hardware suppliers earn 1.66 pp and rival developers 1.61 pp over $[0, +20]$ ($p \approx 0.01$), with abnormal trading volume on the announcement day itself; deployers—firms using AI inside conventional businesses—show a CAR of -0.07 pp with a standard error of 0.25 pp and no abnormal volume at any horizon: a genuine zero, not offsetting disagreement. Third, the two dimensions interact exactly as derived-demand logic implies: capability carries no average price, but each standard deviation of capability adds roughly 1.3 pp to the hardware premium (interaction $p = 0.088$). A more capable model is bigger news for compute suppliers—not for the average “exposed” firm.

The letter’s contribution is to unbundle “AI exposure.” Exposure rankings, whether based on labor-task text measures or investment intensity, conflate firms on opposite sides of the same shock: a chip designer, a cloud platform, and a software deployer are all “highly exposed,” yet their announcement returns differ by economically large margins and even in the sign of the trading response. Once positions are separated, the pricing loads on the compute-complementarity channel and on positive industry spillovers to rivals—the contagion pattern of Lang and Stulz (1992) rather than the competitive-displacement pattern documented for conventional product introductions (Chen et al., 2005)—while the displacement channel that dominates public commentary is priced at a precise zero. A companion full-length paper documents *when* this pricing rule formed (it is a 2025 phenomenon, coinciding with the reasoning paradigm’s shift of compute demand toward inference) and reports a forensic measurement audit in which two opposite headline results from an earlier, hand-collected vintage of the data prove to

be artifacts of misdated events and contaminated return windows. Here we hold the reproducible pipeline fixed and focus on the cross-sectional question: capability or position.

Section 2 describes the data. Section 3 presents the capability tests, the position premia, and the volume evidence. Section 4 summarizes robustness. Section 5 concludes.

2. Data

2.1. Events and dates

The event universe intersects two independent public sources: the AI Timeline chronology of model releases and Artificial Analysis (AA) benchmark coverage. Requiring AA coverage screens out announcements too minor to be benchmarked; identity matching between the sources is resolved by a scored matcher with every manual decision recorded in an auditable table, and same-day family announcements are merged into single events. Of 134 candidate events, 125 have day-level dates verified against first-party evidence and 124 enter the main sample (one event is excluded because its identity is inconsistent across sources): 3 events from 2022, 8 from 2023, 45 from 2024, 59 from 2025, and 9 from 2026. Eighty-seven are language-model releases with AA intelligence scores; thirty-five are media-generation releases; forty-seven release open weights; forty-one are reasoning models.

Two dating pitfalls deserve note because they generalize to any AI event study. Vendor content systems sometimes render timestamps in UTC, shifting a U.S. afternoon announcement to the next calendar day; and Chinese developers' pages carry Beijing dates, which typically coincide with the cor-

rect U.S. reaction day rather than requiring adjustment. We additionally recover intraday release times for 55 events from page metadata and archived first captures; only two events show positive evidence of after-hours release, and Section 4 confirms that no feasible re-dating changes the results.

2.2. Positions, capability, and outcomes

Each of the 108×25 (firm, developer) pairs is coded on eight dimensions under a fixed codebook by two independent coders; disagreements (1.0% of cells) were adjudicated against the codebook with all decisions logged. Event-level indicators take the union across the event’s developers. In the main Nasdaq-100 sample, 25% of firm-events carry a hardware relationship, 5% cloud, 12% integrator, 20% deployer, and 5% competitor; 26% belong to control firms with no relationship on any dimension. Capability is the AA intelligence index of the event’s flagship model (the highest-scoring variant; sample standard deviation 12.7). Because the index is a composite over standardized benchmarks, it is comparable across developers at a point in time—precisely the comparison our within-year specifications exploit.

Daily total returns come from adjusted closes (January 2021 to April 2026). Market-model CARs use the S&P 500 as market proxy with estimation window $[-200, -10]$ (minimum 120 observations), cumulated over $[0, +w]$ for $w \leq 20$ and a pre-event window $[-10, -2]$; we deliberately avoid a Nasdaq-100 benchmark because sample firms are its constituents, and we compute three-factor CARs for robustness. Abnormal volume is the deviation of daily log volume from its estimation-window mean, in estimation-window standard deviations, averaged over the event window. Firm controls (log assets, book-to-market with a negative-equity flag, momentum, volatility) are

matched under a strict 45-day as-of rule so that only statements publicly available at the event enter. Standard errors are clustered by event (CR2) throughout; event-fixed-effects specifications identify from within-event variation alone.

3. Results

3.1. *Capability is not priced*

Table 1 regresses $CAR[0, +20]$ on capability with firm controls and year fixed effects, so the slope compares releases within the same year—not GPT-4 against successors three benchmark generations later. Column (1): the pooled slope is -0.0001 per index point ($p = 0.51$). Column (2) restricts to the fifty closed-source releases, where appropriability logic predicts the steepest capability rents; the slope is -0.0006 ($p = 0.024$). Column (3): among the thirty-seven open-weight releases the slope is 0.0000 ($p = 0.93$). Column (4) is the placebo: among firms with no ecosystem relationship the slope is -0.0001 ($p = 0.80$). Capability scores do not predict returns where no economic channel exists, so the closed-source negative is not a mechanical artifact of the score itself.

The negative closed-source slope has a natural reading: anticipation. Within a year, the most capable releases are flagship events, rumored for months and often pre-announced to the day, while lower-scoring releases arrive with less warning and deliver larger surprises. Three facts support this reading. The eight highest-scoring closed releases in the sample all carry positive average CARs of $+0.1$ to $+2.9\%$ —flagships are not punished; they are pre-priced. The negative slope concentrates in the later sample (-0.0006 ,

Table 1: Model capability and abnormal returns

	(1) All	(2) Closed	(3) Open	(4) Placebo (unrelated firms)	(5) Interaction
AA intelligence	−0.0001 (0.0002)	−0.0006** (0.0002)	0.0000 (0.0003)	−0.0001 (0.0004)	−0.0003 (0.0002)
× Upstream hardware					0.0010* (0.0006)
Upstream hardware					0.0207*** (0.0078)
Events	87	50	37	87	87
Observations	8,562	4,917	3,645	2,784	8,562

Notes: Dependent variable is the market-model CAR[0, +20]. All columns include firm controls (log assets, book-to-market with negative-equity flag, volatility, momentum) and year fixed effects; capability is centered in column (5). Standard errors clustered by event (CR2). *, **, *** denote 10%, 5%, 1%.

$p = 0.04$, in 2025–2026 against -0.0001 , $p = 0.87$, in 2022–2024), when release calendars became crowded and heavily previewed. And replacing the level with a surprise measure—the score’s distance above the pre-event frontier maximum—yields the same pattern, because benchmark-measured surprise still correlates with exactly the flagship status that markets anticipate; separating true expectation errors from levels would require ex-ante forecast data that do not yet exist for model benchmarks.

Column (5) shows what capability *does* do: interacted with the hardware indicator (capability centered), each index point adds 0.10 pp to the hardware premium ($p = 0.088$)—roughly 1.3 pp per standard deviation—while

the hardware main effect at mean capability is 2.07 pp ($p = 0.010$). Capability is priced as a scaling factor on derived compute demand, not as a stand-alone characteristic.

3.2. Position is priced

Table 2 turns to positions. Hardware suppliers earn 1.66 pp more than unrelated constituents over the twenty days after a release ($p = 0.012$); the estimate is 1.79 pp under the three-factor benchmark and 1.54 pp with event fixed effects, where identification comes only from within-event comparisons of firms facing the same release, the same market conditions, and the same AI sentiment. Rival model developers—listed firms that develop competing frontier models—earn a statistically indistinguishable 1.61 pp. The positive competitor spillover is itself informative: product-introduction studies typically find negative wealth effects for rivals (Chen et al., 2005); the positive sign indicates that in this period a release conveyed industry-level growth information that dominated competitive displacement, the contagion pattern of Lang and Stulz (1992) in a fast-expanding technology sector.

The deployer coefficient is -0.07 pp with a standard error of 0.25 pp. This is not an underpowered non-result: the 95% confidence interval, $[-0.56, +0.42]$ pp, excludes deployer losses larger than one third of the hardware gain. Whatever the long-run effects of foundation models on downstream businesses, four years of release events contain no evidence that markets price displacement on announcement. The remaining positions sharpen the picture. Cloud platforms—the most direct resellers of model capability—show a price response of zero (-0.11 pp, $p = 0.89$), a fact whose meaning emerges from the volume evidence below. Integrators are mildly negative and impre-

Table 2: Ecosystem position and abnormal returns

Dependent variable:	(1) $CAR^{MM}[0, 20]$	(2) $CAR^{FF3}[0, 20]$	(3) $CAR^{MM}[0, 20]$
Upstream hardware	0.0166** (0.0065)	0.0179*** (0.0065)	0.0154** (0.0066)
Upstream cloud	-0.0011 (0.0078)	0.0020 (0.0074)	-0.0018 (0.0076)
Downstream integrator	-0.0051 (0.0049)	-0.0041 (0.0048)	-0.0053 (0.0050)
Downstream deployer	-0.0007 (0.0025)	0.0002 (0.0024)	-0.0011 (0.0025)
Competitor	0.0161** (0.0064)	0.0159*** (0.0058)	0.0158** (0.0064)
Investor, owner	Yes	Yes	Yes
Firm controls	Yes	Yes	Yes
Year fixed effects	Yes	Yes	—
Event fixed effects	—	—	Yes
Observations	12,112	12,112	12,112
Events	124	124	124

Notes: Investor and owner/publisher indicators included (positive, insignificant; their positive cells concentrate in few developers, hence few clusters). Standard errors clustered by event (CR2). *, **, *** denote 10%, 5%, 1%.

cise (-0.51 pp, $p = 0.30$). The economically meaningful contrast is the roughly 1.7 pp spread between hardware and deployers within the same event window—firms distinguished only by where they sit in the production chain.

The premium is not a single-stock story. Dropping any one hardware firm

leaves the coefficient between 1.2 and 1.6 pp across twenty-three leave-one-out fits; excluding NVIDIA specifically gives 1.5 pp ($p = 0.023$); and hardware members outside the semiconductor index proper (server, networking, and storage suppliers) earn at least as much as semiconductor names. Nor is it a level effect of the AI era: at the portfolio level, a calendar-time strategy long the hardware suppliers and short the downstream firms earns a three-factor alpha of roughly 24% annualized over the sample ($p = 0.054$), though with release windows covering 56% of trading days we read the portfolio as economic scale rather than identification, which rests on the within-event estimates.

Figure 1 traces the accumulation. In the full sample the hardware premium is invisible at the close of the announcement day, emerges by $[0, +10]$ (0.98 pp, $p = 0.02$), and reaches 1.66 pp by day twenty; in the 2025–2026 subsample—after the pricing rule strengthened, as the companion paper documents—it is significant from $[0, +2]$ (+0.51 pp) and accumulates monotonically to 4.28 pp. The deployer line never leaves zero at any horizon. Two checks address the concern that slow accumulation reflects confounded drift: controlling for each firm’s own pre-event CAR leaves the post-event estimates essentially unchanged (4.28 to 4.35 pp in the late sample), and the pre-event hardware run-up that does exist (+1.4 pp over $[-10, -2]$ in 2025–2026) is not accompanied by any abnormal volume, consistent with thin pre-announcement positioning rather than information leakage.

3.3. Volume separates attention, disagreement, and irrelevance

Abnormal volume completes the taxonomy (Table 3), and it is where the three downstream fates separate most cleanly. Hardware suppliers trade 0.12

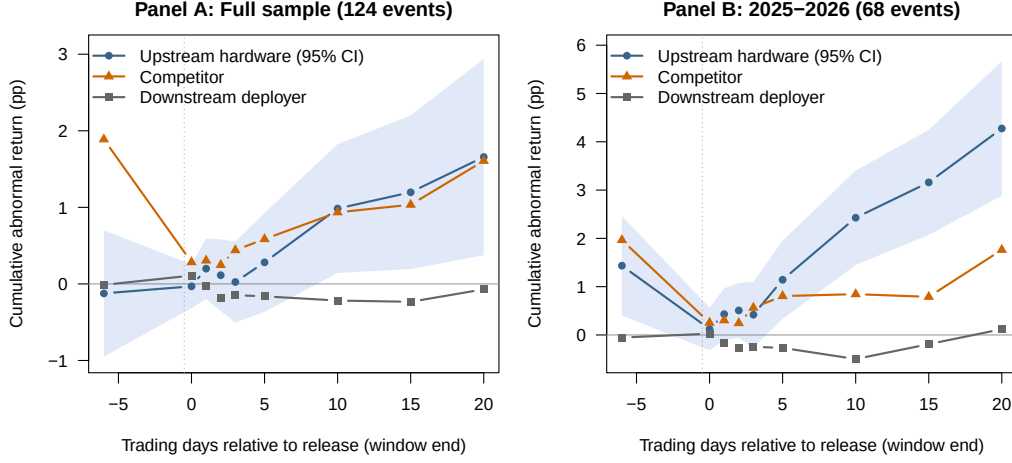


Figure 1: Cumulative abnormal returns by ecosystem position and accumulation window. Coefficients from window-by-window cross-sectional regressions with firm controls and year fixed effects; shading is the 95% band for the hardware coefficient (CR2, clustered by event). The leftmost point is the pre-event window $[-10, -2]$. Left panel: full sample; right panel: 2025–2026.

standard deviations above normal on the announcement day ($p < 0.001$)—including in the full sample, where the same-horizon price response is still indistinguishable from zero. Attention arrives immediately; directional consensus takes days to form. That sequencing—volume first, price later—is the signature of gradual information incorporation rather than of a return drift unrelated to the event.

Deployers show no abnormal volume at any horizon ($p = 0.74$ at $[0, +1]$). Combined with the precise price zero, this rules out the interpretation that deployer investors disagree and net out: there is simply nothing to trade. Cloud platforms are the mirror image and the most striking cell in the panel: abnormal volume of 0.29 standard deviations at $[0, +1]$ (0.44 in 2025–2026),

Table 3: Abnormal trading volume

Window:	Full sample		2025–2026	
	$[-10, -2]$	$[0, +1]$	$[-10, -2]$	$[0, +1]$
Upstream hardware	0.033 (0.026)	0.121*** (0.036)	0.046 (0.032)	0.106*** (0.036)
Upstream cloud	0.204*** (0.041)	0.292*** (0.049)	0.314*** (0.046)	0.435*** (0.063)
Downstream deployer	−0.019 (0.015)	0.006 (0.019)	0.024 (0.018)	0.028 (0.021)
Competitor	−0.037 (0.029)	−0.017 (0.043)	−0.014 (0.041)	−0.081 (0.052)
Event fixed effects	Yes	Yes	Yes	Yes

Notes: Standardized abnormal log volume (deviation from the $[-200, -10]$ mean in estimation-window SDs, averaged over the window). Firm controls included; SEs clustered by event (CR2). *, **, *** denote 10%, 5%, 1%.

already 0.20 standard deviations in the *pre-event* window, and yet a price response of zero throughout. In disagreement models, volume without price movement is the fingerprint of investors interpreting the same public signal with opposite priors (Kandel and Pearson, 1995). Cloud platforms fit naturally: a frontier release is simultaneously good news for their compute-rental business and bad news for their own model ambitions and gross margins, and the market splits on the net. The taxonomy, then: hardware is priced with consensus, cloud is traded without consensus, and deployers are ignored on both margins.

4. Robustness

The position premia are significant in 16–17 of 20 specifications spanning benchmark models (S&P 500, Nasdaq-100, SOX, three-factor), accumulation windows ($[0, 10]$, $[0, 15]$), winsorization at 1%/99%, restriction to high-confidence dates, exclusion of the six multi-component events, event fixed effects, and a pre-event CAR control; the deployer zero and the null hardware $\times$ open-weight interaction ($p = 0.90$) hold in all twenty. The open-weight null deserves emphasis given the public narrative that efficient open models threaten compute rents: releases that publish weights generate the same hardware premium as proprietary ones. Event-day conventions do not matter: announcement timestamps recovered for 55 events identify only two after-hours releases, and shifting *every* event by one trading day leaves the late-sample hardware estimate at 4.7 pp ($p < 0.001$). The capability results are likewise insensitive to three-factor benchmarks, to the surprise transformation, and to restricting the sample to firms with any AI relationship; and an independent 3,144-specification search over outcomes, samples, moderators, and fixed-effect structures finds the hardware–time interaction positive in 100% of specifications (family-wise $q = 0.002$).

5. Conclusion

Around LLM releases, equity markets do not pay for capability—they pay for position. Benchmark scores carry no positive premium, and a negative one among the anticipated flagships where capability rents should be steepest; hardware suppliers and rival developers revalue by 1.6–1.7 pp per release while AI-deploying firms show precisely zero price and volume response; and

capability enters pricing only by scaling the hardware premium. The volume evidence adds a taxonomy that returns alone cannot deliver: consensus pricing upstream, unresolved disagreement at cloud platforms, and genuine irrelevance downstream.

The results carry two lessons. For research that ranks firms by AI exposure, exposure conflates positions with opposite pricing; the economically active margin is the supply-chain role, and pooling roles attenuates estimated AI effects toward zero. For the public debate, four years of high-frequency technology news contain no evidence that markets price AI displacement of downstream businesses on announcement—the displacement narrative, whatever its long-run merit, is not in prices. A natural extension is expectations data: analyst reports and earnings-call language can date when compute-demand reasoning entered the sell-side lexicon and test the anticipation mechanism we infer from the flagship gradient directly.

Data availability

The event pipeline, coding decision tables, and estimation code are available from the authors.

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